DAURES GREEN HYDROGEN CONSORTIUM
DAURES GREEN HYDROGEN VILLAGE
University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use
With funding from the:
Federal Ministry of Research, Technology and Space

Macroeconomic results

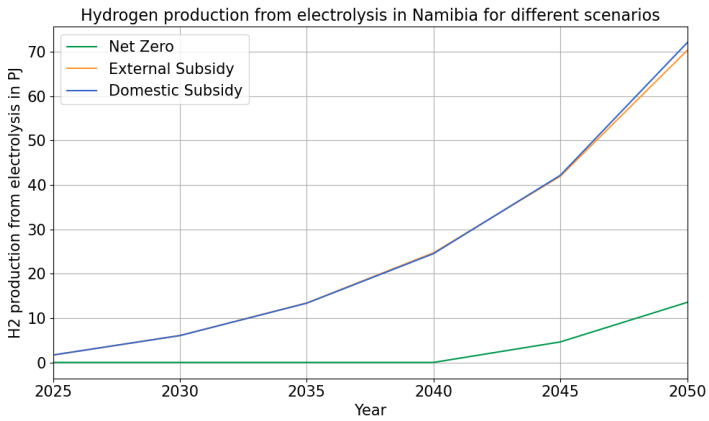
Scenarios

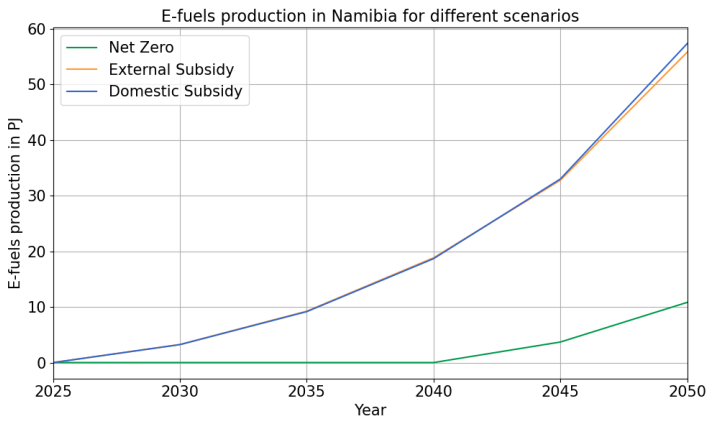
Scenario	Net Zero	External Subsidy	Domestic Subsidy
Climate policy	World: IEA WEO 2024 „Announced pledges"; Namibia: NDC scenario		
Hydrogen subsidy	-	3.0	3.0
Subsidy financing	-	Foreign Government	Namibian Government

Hydrogen and synthetic fuels

A subsidy on hydrogen production has a direct production effect, not only on the production of hydrogen but also on the production of synthetic fuels since hydrogen is an important input for synthetic fuels production.

Production



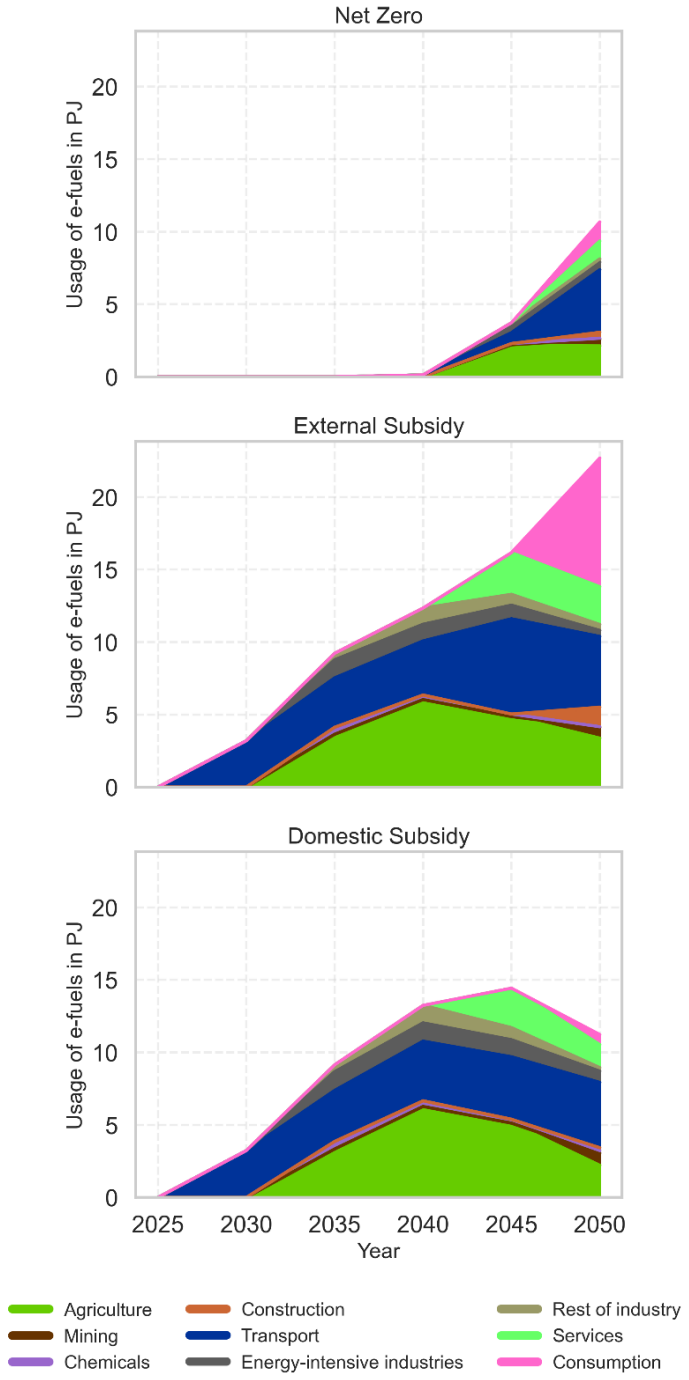


Compared to the reference scenario "Net Zero" (no subsidy on H2 production) the production of hydrogen and e-fuels until 2050 increases when a subsidy is paid for hydrogen production. Green hydrogen production increases from 14 PJ in 2050 to around 70 PJ in 2050, while e-fuels production increases from roughly 11 PJ in 2050 to around 56 PJ in 2050. It is noticeable, that hydrogen and e-fuels production increase simultaneously because the hydrogen produced is mostly used for e-fuels production further on. Synthetic gas production remains below 1.6 PJ for all scenarios and years and, hence, does not seem to play a significant role in the Namibian energy system.

Usage

Since most of the produced hydrogen flows into e-fuels production, direct usage in final energy consumption remains below 2 PJ for all scenarios and years. E-fuels usage concentrates on the agriculture and transport sector and, in the "External Subsidy" scenario the 2050 production peak is used by household consumption (e.g. private transport, heating).

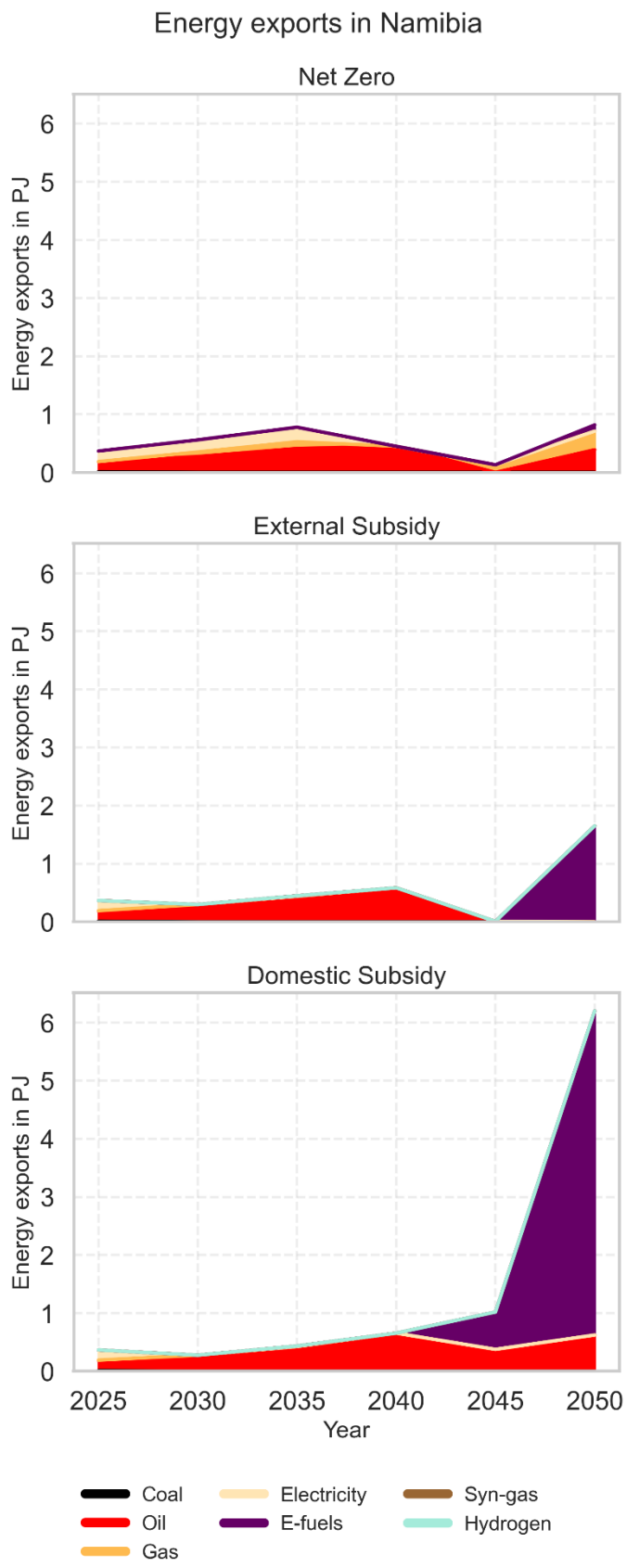
E-fuels usage for final energy consumption in Namibia

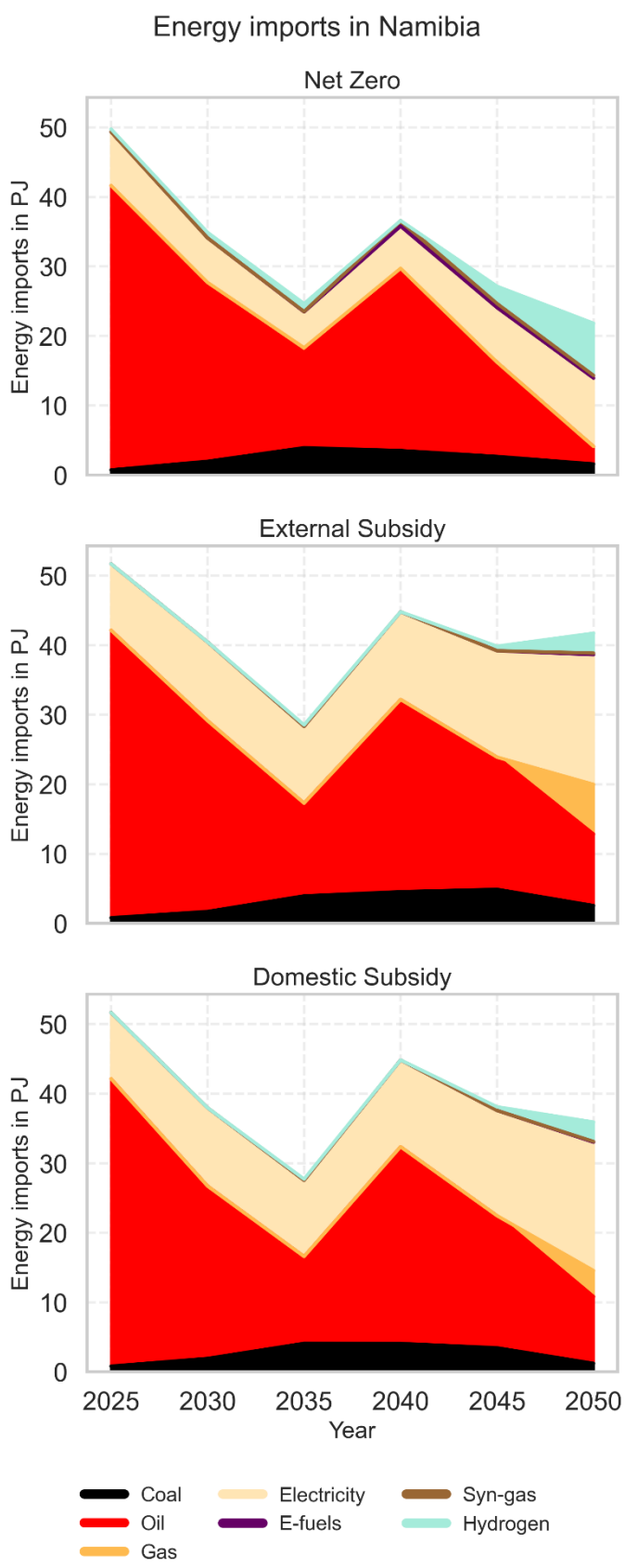


Energy trade

Currently, the Namibian economy is rather dependent of oil imports and the green hydrogen strategy is strongly oriented in export of hydrogen and/or derivatives. The scenario analysis reveals that, without further policies, export remains rather low throughout all scenarios.

Energy import dependency, on the other hand, differs between sectors. While oil imports drastically decrease until 2050, electricity imports increase, especially when hydrogen production is subsidized. This is due to the large amounts of electricity needed for hydrogen and e-fuels production. Interestingly, small amounts of hydrogen are imported even in the subsidy scenarios.





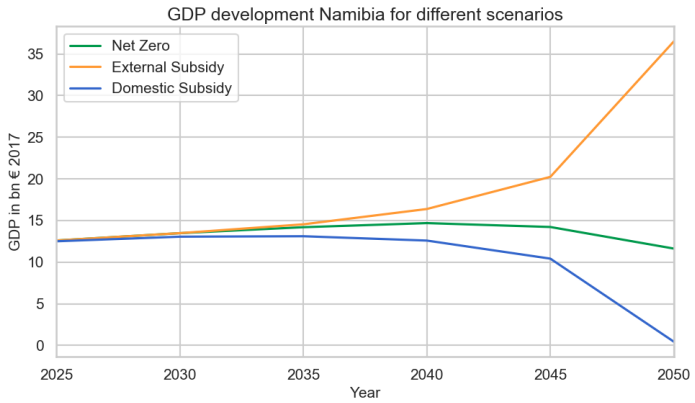
Macroeconomic results

Economy

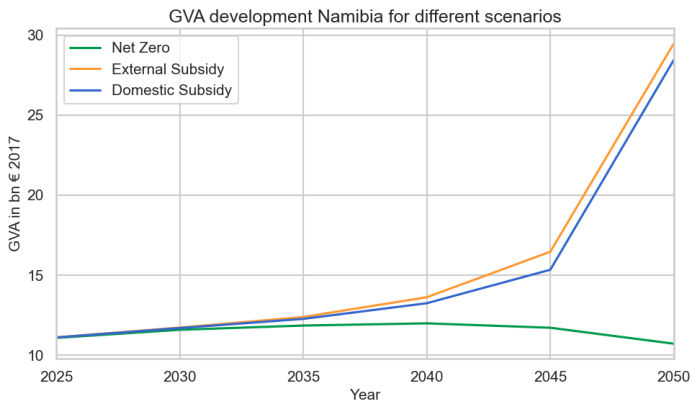
The transformation towards a green hydrogen economy offers a chance for Namibia to reshape its economy and strengthen it for the future. The macroeconomic effects of such a transformation are manifold.

Economic growth and development:

Gross domestic product (GDP) is the most common indicator to analyze effects on economic growth. The scenario analysis clearly shows that the economy profits from a subsidy which is paid by foreign governments ("External Subsidy"). On the other hand, the Namibian economy deeply suffers from a subsidy paid by the Namibian government itself ("Domestic Subsidy"). This loss in welfare is due to significant reductions in consumption and investment, in order to enable the subsidy payments to production entities.

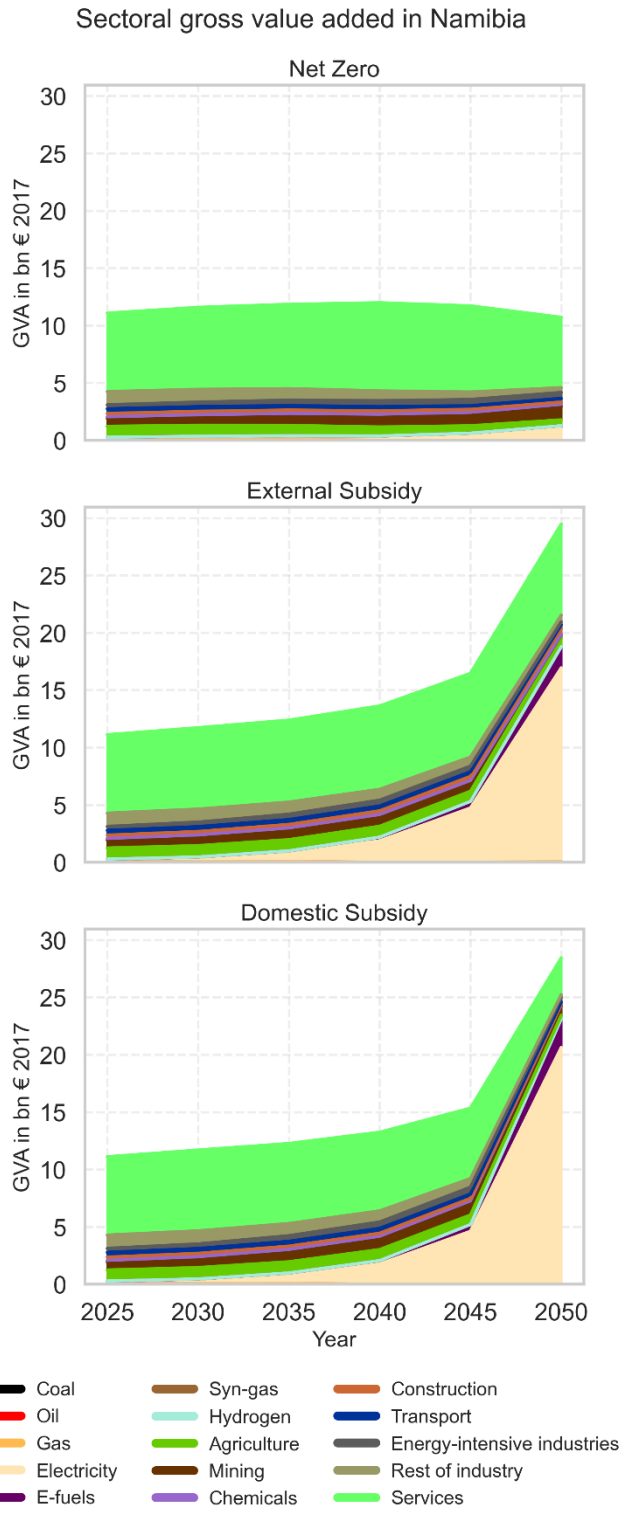


On the production side, however, the source for subsidy financing does not significantly impact the development of gross value added (GVA) in industrial production. Both counterfactual scenarios, "External Subsidy" and "Domestic Subsidy", show large increases in total GVA.



Industrial sectors:

When looking closer on industry sectors and how such a transformation might restructure the Namibian economy, results show, that especially the electricity sector increases strongly.



Policy recommendations

A subsidy will foster hydrogen production. However, the Namibian target of around 1300 PJ seems rather ambitious. In addition, Namibia should create attractive funding conditions in order to attract foreign investments. Financing a hydrogen subsidy on its own does not seem recommendable.

Contact

Claudia Hofer, M. Sc. (Email: claudia.hofer@ier.uni-stuttgart.de)

Institute of Energy Economics and Rational Energy Use, University of Stuttgart

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